Digital Electronics Laboratory

Laboratory Exercise 3. Basic Adders

Objectives:

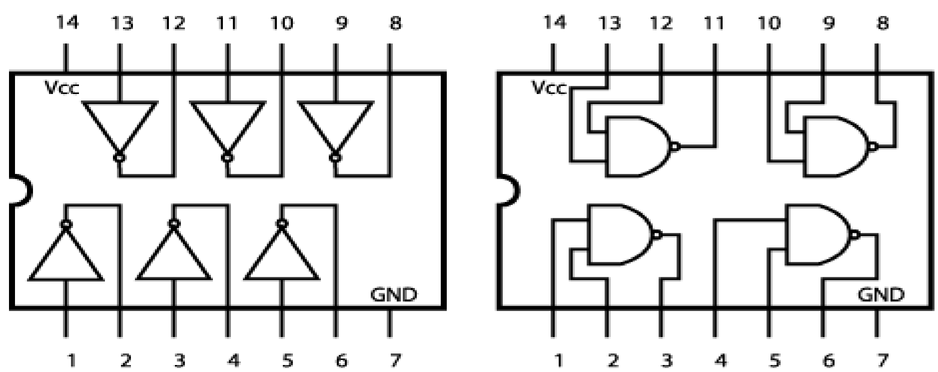
- To implement a half adder
- To implement a full adder
- To implement a four bit parallel adder

Pre-Laboratory

There are two tasks that you must perform **prior** to sitting this laboratory:

- Read the laboratory assignment in full.
- Draw up the truth tables for a half and full adder.

Useful Chip Diagrams:



7404(NOT) **7400(NAND)**

The Laboratory:

Section 1. The Half Adder

(a) Construct the circuit as shown in Figure 1.

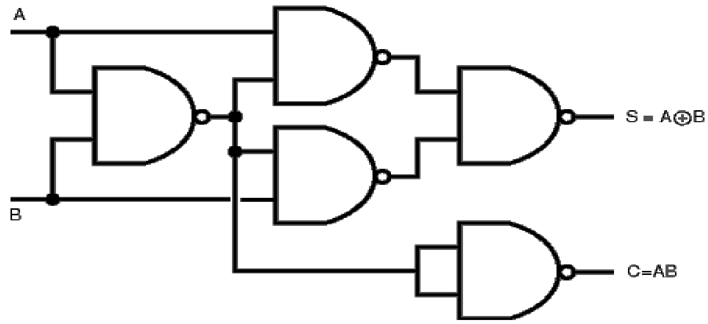


Figure 1. The Half adder circuit

(b) Vary the inputs A and B to obtain all the possible combinations and complete a Truth table for the sum output S and the carry output C.

(c) Is this as you expected? Why is the half adder not suitable for all addition operations? (i.e. why do we need a full adder?)

Section 2. The Full Adder

(a) Build on your previous circuit to create the circuit as shown in Figure 2.

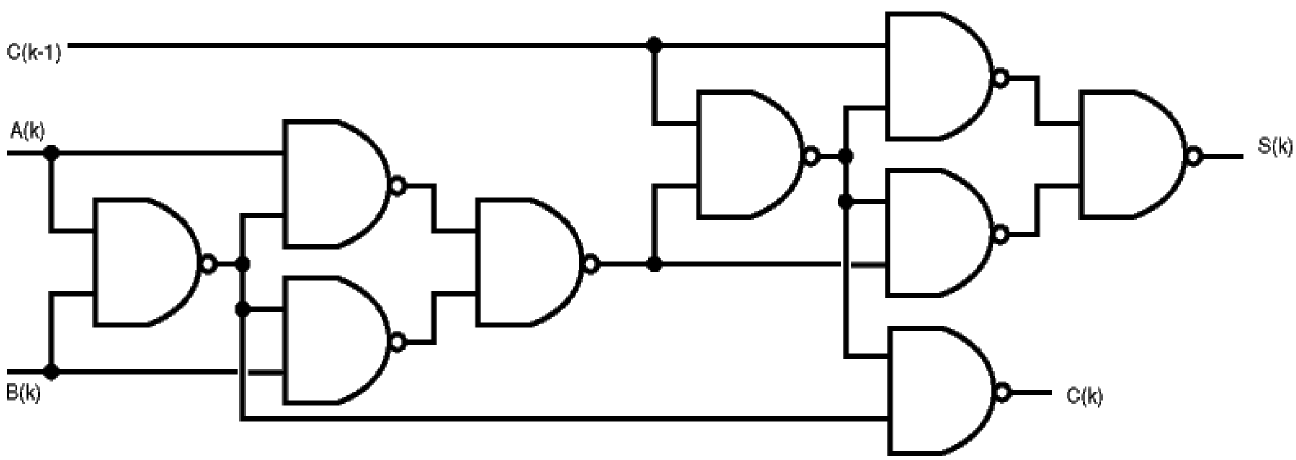


Figure 2. The Full Adder Circuit

(b) Vary inputs A_k B_k and C_{k-1} and complete a Truth table for the circuit.

Section 3. The Four Bit Adder

(a) The 74LS83a is a high speed 4-bit Full Adder. It accepts two 4-bit words (A_1 to A_4) and (B_1 to B_4) and a carry input (C_0). It generates the binary sum outputs (S_1 to S_4) and the carry output (C_4) from the most significant bit. Connect the two binary numbers shown below to the inputs of the 4-bit parallel adder. Connect the carry input C_0 to 0.

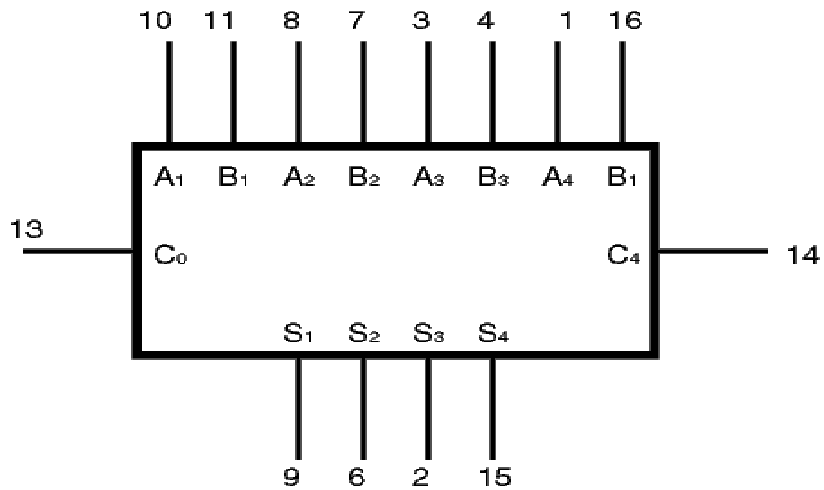


Figure 3. The Four Bit Adder

Check the operation of the Four Bit adder by adding the numbers:

(a) 0110 + 0101 (b) 1011 + 0011 (c) 1110 + 0100 (d) 1111 + 1111

Section 4. Conclusions

(a) State briefly, but clearly, what you have gained from this laboratory.